# ITA03 – MOBILE COMPUTING

# ASSIGNMENT 3 – CO3

| STUDENT DETAILS | |
|---|---|
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| Course | ITA03 – Mobile Computing |
| Assignment | Assignment 3 – CO3 |

Q1 – WIRELESS NETWORK ARCHITECTURE AND MOBILITY ANALYSIS

## 1.1 Introduction to wireless networks

A wireless network is a communication system in which devices exchange data through radio or other wireless signals. It removes the need for a dedicated wired connection between each mobile device and the network. Wireless networks are useful where users move frequently, where cable installation is difficult, or where temporary connectivity is required.

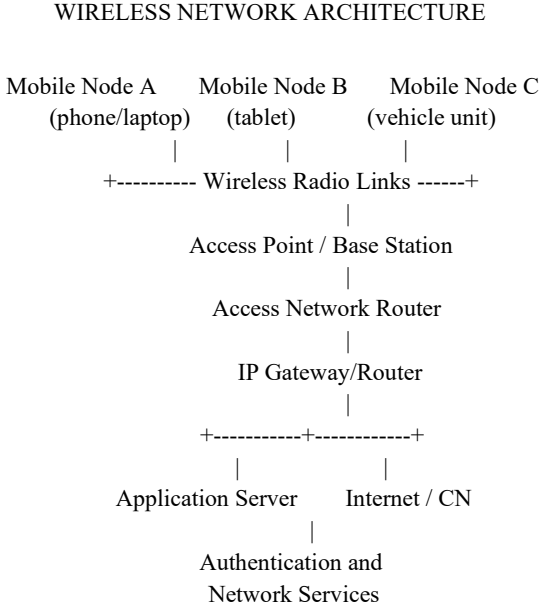
A wireless network normally contains mobile nodes, wireless access points or base stations, communication links, gateways and network services. Mobile nodes generate and receive application data. Access points and base stations provide radio access to an infrastructure network. Gateways connect the wireless segment to other IP networks or the Internet.

## 1.2 Characteristics and components

| **Component** | **Role in the network** |
|---|---|
| Mobile node | A smartphone, tablet, laptop, ambulance terminal or other device that communicates while moving. |
| Wireless access point / base station | Provides radio connectivity and connects wireless users to the network infrastructure. |
| Wireless link | Radio communication path between a mobile node and an access point, base station or neighbouring node. |
| Gateway / router | Forwards packets between the wireless access network and other networks. |
| Mobility management entity | Maintains or coordinates location and attachment information when a device changes networks. |
| Correspondent node | The remote endpoint communicating with a mobile node. |

## 1.3 Proposed wireless network architecture

A suitable general architecture for a university or township environment contains mobile users connected to nearby access points or cellular base stations. These access nodes connect to an aggregation router and then to a gateway. The gateway provides access to application servers and external networks. When a mobile node moves between coverage areas, the network detects the new point of attachment and performs the required handoff or mobility update.




## 1.4 Wireless communication characteristics

- Limited and shared bandwidth: multiple devices use the available radio resources, so throughput depends on channel allocation and network load.
- Variable signal strength: distance, obstacles, interference and device movement affect received signal quality.
- Variable connectivity: a node may temporarily lose contact when it moves outside coverage or experiences obstruction.
- Mobility: a device may change its access point, base station or neighbouring-node set while a session is active.
- Wireless errors: fading, interference and collisions can cause packet corruption or retransmission.
- Security exposure: radio signals can be received beyond the physical boundary of a building, so authentication and encryption are important.

## 1.5 Effect of limited bandwidth and connectivity

Wireless bandwidth is a shared resource. When the number of active users increases, each user may receive a smaller share of the available capacity. Congestion can increase queuing delay, packet loss and retransmissions. A network designer must therefore consider the expected number of users, application data rate, traffic burstiness and quality-of-service requirements.

Connectivity can be interrupted by movement, weak signal strength, interference, access-point failure or network congestion. During a handoff, the mobile node may need to discover a new access point,

authenticate, obtain or confirm network configuration and update mobility information. The interruption duration depends on the technology and the handoff procedure.

## 1.6 Communication flow

1. The mobile node scans for an available wireless network.
2. It selects a suitable access point or base station based on network policy and radio conditions.
3. The node authenticates and establishes a wireless link.
4. The access network provides IP connectivity and a route to the gateway.
5. The mobile node sends application packets through the access point, router and gateway.
6. The gateway forwards packets toward the correspondent node or application server.
7. Return packets travel through the network toward the current attachment point.
8. If the node moves, the network detects the change and performs handoff or mobility management.

## 1.7 Small wireless communication scenario

Consider a student carrying a smartphone from an academic block to a library. Initially, the phone is connected to Access Point AP1. As the student approaches the library, the signal from AP1 becomes weaker and AP2 becomes more suitable. The phone discovers AP2, authenticates and changes its point of attachment. If the network supports seamless mobility, ongoing communication can continue with a short interruption.

## 1.8 Mobility scenario and event sequence

| **Stage** | **Network event** | **Expected communication behaviour** |
|---|---|---|
| 1 | Mobile node connected to AP1 | Packets are delivered through AP1. |
| 2 | Signal from AP1 decreases | Packet errors or delay may begin to increase. |
| 3 | AP2 is detected | The node evaluates signal quality and network availability. |
| 4 | Handoff begins | Some packets may be delayed, buffered or lost. |
| 5 | Attachment changes to AP2 | The node resumes transmission through AP2. |
| 6 | Route or mobility state is updated | Communication continues using the new attachment. |

## 1.9 Mobility decision sequence – pseudocode

```
ALGORITHM SelectWirelessAttachment(MN, CandidateAPs)

INPUT:
  MN – mobile node
  CandidateAPs – detected access points and their signal quality
```

```
OUTPUT:
  Selected access point or NO_SUITABLE_AP

BEGIN
  SelectedAP <- NULL
  BestSignal <- MIN_SIGNAL

  FOR each AP in CandidateAPs DO
    IF AP.Available = TRUE AND AP.SignalQuality > BestSignal THEN
      BestSignal <- AP.SignalQuality
      SelectedAP <- AP
    END IF
  END FOR

  IF SelectedAP = NULL THEN
    RETURN NO_SUITABLE_AP
  END IF

  IF SelectedAP != MN.CurrentAP THEN
    INITIATE HANDOFF TO SelectedAP
    WAIT FOR AuthenticationResult
    IF AuthenticationResult = SUCCESS THEN
      UPDATE MN.CurrentAP
      RETURN HANDOFF_SUCCESS
    ELSE
      RETURN HANDOFF_FAILURE
    END IF
  END IF

  RETURN CURRENT_ATTACHMENT_RETAINED
END
```

## 1.10 Mobility performance analysis

Packet loss may occur when packets are transmitted during a handoff and the old link is no longer usable. Delay may increase because packets are buffered, retransmitted or routed through a temporary path. Increasing mobility speed can cause more frequent handoffs and reduce the time available for stable communication with each access point.

A successful communication case occurs when the new access point is detected early, authentication completes quickly and routing or mobility state is updated before the old connection becomes unusable. An interrupted case occurs when the old link fails before the new link is ready, or when authentication and route updates take too long.

## 1.11 Analytical evidence and metrics

| **Metric** | **Definition / measurement method** |
|---|---|
| Handoff delay | Time from handoff initiation to successful data transfer through the new attachment. |
| Packet delivery ratio | Packets received successfully divided by packets generated, multiplied by 100. |

| | |
|---|---|
| Packet loss ratio | Packets not received divided by packets generated, multiplied by 100. |
| End-to-end delay | Time between packet generation at source and reception at destination. |
| Throughput | Successfully delivered data bits per unit time. |
| Connection interruption | Duration during which the mobile node cannot exchange data. |

Student A's primary responsibility is to explain the architecture, components, wireless characteristics and infrastructure communication. Student B's primary responsibility is to analyse mobility scenarios, handoff interruption, packet loss, delay and changing network conditions. Both students must be able to explain the complete communication sequence.

Q2 – MOBILE IP AND MOBILITY MANAGEMENT

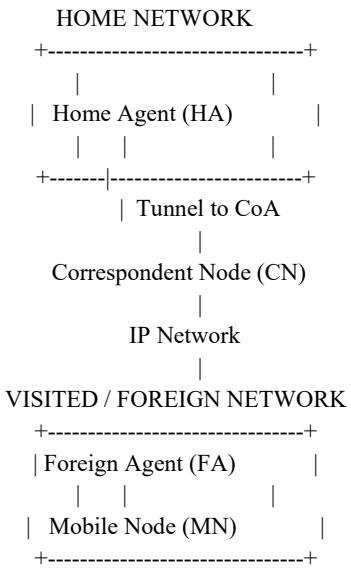
## 2.1 Introduction to Mobile IP

Mobile IP is a mobility-management approach that allows a mobile node to move from its home network to a foreign or visited network while retaining its permanent home address. Without mobility support, changing networks may change the device's IP address and interrupt sessions that depend on the original address. Mobile IP maintains a binding between the home address and a care-of address representing the mobile node's current location.

## 2.2 Mobile IP entities

| **Entity** | **Function** |
|---|---|
| Mobile Node (MN) | The moving device that retains its home address while changing its point of attachment. |
| Home Agent (HA) | A router on the home network that maintains the mobile node's current care-of address and forwards packets toward it. |
| Foreign Agent (FA) | In traditional Mobile IPv4, an agent in the visited network that assists the mobile node with care-of address and packet delivery. |
| Correspondent Node (CN) | The remote device or server communicating with the mobile node. |
| Home Address | Permanent IP address assigned to the mobile node on its home network. |
| Care-of Address (CoA) | Address representing the mobile node's current location in a visited network. |

## 2.3 Mobile IP architecture




## 2.4 Role of the Mobile Node

The mobile node is the device whose network attachment changes. It stores its home address and discovers the network in which it is currently located. When it enters a visited network, it obtains or discovers a care-of address and registers that address with its home agent. It continues to use its home address as its stable identity for communication.

## 2.5 Role of the Home Agent

The home agent maintains a mobility binding that maps the mobile node's home address to its current care-of address. When a correspondent node sends a packet to the home address, the packet reaches the home network. The home agent intercepts or receives the packet, encapsulates it and forwards it through a tunnel toward the care-of address.

## 2.6 Role of the Foreign Agent

In traditional Mobile IPv4, the foreign agent advertises its availability, assists the mobile node in obtaining a care-of address and may act as the endpoint of the tunnel. It can decapsulate packets and deliver the original packets to the mobile node. The exact role depends on whether the mobile node uses a foreign-agent care-of address or a co-located care-of address.

## 2.7 Role of the Correspondent Node

The correspondent node is the communication endpoint that sends packets to or receives packets from the mobile node. It may not need to know the mobile node's current physical location because it continues to address packets to the mobile node's home address. The home agent performs forwarding toward the current care-of address.

## 2.8 Agent discovery

A mobile node must determine whether it is on its home network or a visited network. Mobility agents may send advertisements. If the node does not receive an advertisement, it can send an agent solicitation and wait for a response. The advertisement identifies the network and the type of agent.

```
ALGORITHM MobileIPAgentDiscovery(MN)

INPUT:
  MN – mobile node

OUTPUT:
  Detected agent or AGENT_NOT_FOUND

BEGIN
  LISTEN for AgentAdvertisement

  IF Advertisement received THEN
    Extract NetworkAddress
    Extract AgentType

    IF AgentType = HOME_AGENT THEN
      Set HomeAgentAddress
    ELSE IF AgentType = FOREIGN_AGENT THEN
      Set ForeignAgentAddress
    END IF

    RETURN AgentType, NetworkAddress
  ELSE
    SEND AgentSolicitation
    WAIT FOR Advertisement

    IF Advertisement received THEN
      RETURN DetectedAgentInformation
    ELSE
      RETURN AGENT_NOT_FOUND
    END IF
  END IF
END
```

## 2.9 Registration and location updating

After discovering a visited network, the mobile node obtains a care-of address and sends a registration request to its home agent. The request includes the home address, care-of address, requested lifetime and an identifier. The home agent authenticates the request and, if accepted, updates the binding. The reply informs the mobile node whether registration succeeded.

```
ALGORITHM RegisterMobileNode(MN, HA, CoA)

INPUT:
  MN – mobile node
  HA – home agent
  CoA – current care-of address
```

```
OUTPUT:
  Registration result

BEGIN
  Create RegistrationRequest
  Request.HomeAddress <- MN.HomeAddress
  Request.CareOfAddress <- CoA
  Request.Lifetime <- RequestedLifetime
  Request.Identifier <- GenerateRegistrationIdentifier()

  Authenticate Request
  SEND Request TO HA
  WAIT FOR RegistrationReply

  IF RegistrationReply.Status = ACCEPTED THEN
    Binding[MN.HomeAddress] <- CoA
    RegistrationExpiry <- CurrentTime + RegistrationReply.Lifetime
    RETURN REGISTRATION_SUCCESS
  ELSE
    RETURN REGISTRATION_FAILURE
  END IF
END
```

## 2.10 Packet delivery and tunnelling

When a correspondent node sends a packet to the mobile node's home address, the packet is routed toward the home network. The home agent checks its binding table. If a valid care-of address exists, the home agent encapsulates the original packet inside a new IP packet addressed to the care-of address. The tunnel carries the encapsulated packet to the visited network. The tunnel endpoint decapsulates it and delivers the original packet to the mobile node.

```
ALGORITHM TunnelPacketToMobileNode(Packet, HomeAddress)

INPUT:
  Packet – original IP packet
  HomeAddress – destination mobile node home address

OUTPUT:
  DELIVERY_SUCCESS or DELIVERY_FAILURE

BEGIN
  IF Binding[HomeAddress] does not exist THEN
    DROP Packet
    RETURN DELIVERY_FAILURE
  END IF

  CoA <- Binding[HomeAddress]
  EncapsulatedPacket <- ENCAPSULATE(Packet, CoA)
  FORWARD EncapsulatedPacket TO CoA

  IF TunnelDeliverySuccessful THEN
    DECAPSULATE at tunnel endpoint
```

```
      DELIVER original Packet TO MobileNode
      RETURN DELIVERY_SUCCESS
    ELSE
      RETURN DELIVERY_FAILURE
    END IF
  END
```

## 2.11 Mobile IP challenges

| Challenge | Explanation | Possible mitigation |
|---|---|---|
| Registration delay | The mobile node must discover the visited network, obtain a CoA and register it. | Use efficient discovery, suitable registration timers and reliable signalling. |
| Tunnelling overhead | Encapsulation adds an outer IP header and consumes extra bandwidth. | Use efficient tunnel management and avoid unnecessary encapsulation where supported. |
| Triangular routing | Packets may travel from CN to HA and then to the mobile node, creating a non-optimal path. | Use route optimisation where supported and secure. |
| Packet loss during mobility | Packets may be sent to an old care-of address while registration is changing. | Buffer packets, update bindings promptly and use retransmission where appropriate. |
| Security | False registration messages could redirect traffic. | Authenticate mobility messages and protect registration exchanges. |
| Scalability | A large number of mobile nodes creates binding and signalling load. | Use efficient state management and scalable agent infrastructure. |

## 2.12 Triangular routing

Triangular routing occurs when packets from a correspondent node first travel to the mobile node's home agent and then are tunnelled to the mobile node's current location. The return path may travel directly from the mobile node to the correspondent node. This can create an inefficient path and increase end-to-end delay. Route optimisation may reduce the detour, but it must preserve authentication and binding security.

## 2.13 Frequent network changes during an active session

If a mobile user frequently changes its network of attachment, the care-of address may change repeatedly. The home agent must receive and process new registrations. Packets may be sent to an old care-of address while the new binding is being installed. This can cause temporary packet loss, additional delay, signalling overhead and possible reordering. Frequent registration also increases processing and security requirements.

- Use prompt agent discovery and avoid unnecessary repeated registration.
- Use suitable registration lifetimes and refresh before expiry.
- Buffer or retransmit packets during short handoff intervals where the system supports it.

- Use secure registration authentication to prevent redirection attacks.
- Use route optimisation where supported to reduce triangular routing delay.
- Coordinate handoff and routing updates so that packets use the new attachment as quickly as possible.

## 2.14 Student responsibilities and joint conclusion

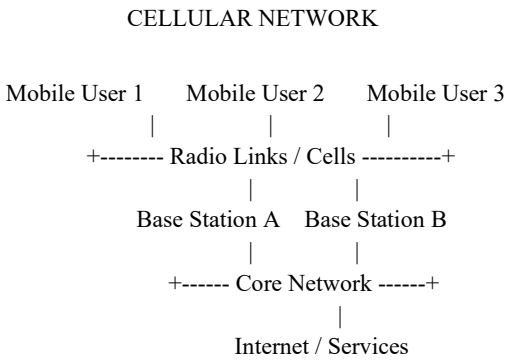
Student A's primary responsibility is to explain the Mobile Node, Home Agent, Foreign Agent, Correspondent Node, registration, location updating, packet delivery and tunnelling. Student B's primary responsibility is to analyse registration delay, tunnelling overhead, triangular routing, packet loss, security and scalability. Both students must be able to defend the complete Mobile IP operation.

Mobile IP maintains communication identity through a stable home address and updates the binding to the current care-of address. Its effectiveness depends on timely discovery, secure registration, reliable tunnelling and efficient handoff. Frequent mobility makes signalling and packet interruption more significant, so mobility management must be coordinated with routing and link-layer handoff.

Q3 – CELLULAR NETWORKS VERSUS AD HOC WIRELESS NETWORKS

## 3.1 Cellular wireless network

A cellular network divides a geographical service area into cells. Each cell is served by a base station that provides radio access to mobile devices. Base stations connect to the operator's core network, which provides authentication, mobility management, routing and access to external networks. The network infrastructure is planned and centrally managed.




## 3.2 Role of base stations and infrastructure

- Provide radio coverage within a cell.
- Coordinate access to radio resources and communication with mobile devices.
- Forward user traffic toward the core network.
- Support handoff when a mobile device moves between cells.
- Enable centralized authentication, mobility management and network monitoring.

## 3.3 Centralised communication and handoff

In a cellular network, a mobile device generally communicates through a base station rather than forwarding traffic for other mobile devices. When a device moves from one cell to another, the network performs a handoff to maintain service. The network can coordinate this process because the base stations and core network are part of a managed infrastructure.

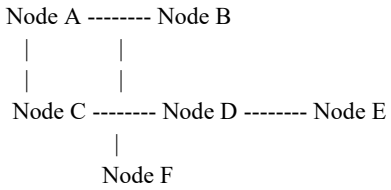
## 3.4 Scalability, coverage and applications

Cellular systems can provide wide-area coverage by deploying many base stations and reusing radio resources across cells. Capacity can be increased by adding sites, sectors or radio resources. Suitable applications include public voice and data service, mobile broadband, fleet connectivity, mobile healthcare and transportation services.

## 3.5 Ad hoc wireless network

An ad hoc wireless network is formed by mobile nodes without requiring fixed base stations or pre-existing network infrastructure. Nodes communicate directly when they are within radio range. If the destination is not directly reachable, intermediate nodes forward packets, creating a multi-hop path. Each node can act as both an end host and a router.

AD HOC NETWORK




Each node can generate, receive and forward packets.
Routes are formed across currently available wireless links.

## 3.6 Dynamic topology and multi-hop communication

The topology of an ad hoc network changes when nodes move, links fail or nodes join and leave. A route that exists at one moment may become invalid later. Routing protocols must therefore discover routes, maintain route state and repair broken paths. Multi-hop forwarding allows nodes outside the direct radio range of the source to communicate through intermediate nodes.

## 3.7 Absence of fixed infrastructure

An ad hoc network can be deployed quickly in an area where fixed infrastructure is unavailable or damaged. It does not depend on a base station for every communication. However, the network depends on participating nodes for forwarding, so node energy, radio range, density and willingness or ability to relay traffic affect performance.

## 3.8 Technical comparison

| Parameter | Cellular Network | Ad Hoc Network |
|---|---|---|

| | | |
|---|---|---|
| Infrastructure | Requires base stations, access infrastructure and core network. | No fixed infrastructure is required. |
| Communication | Mobile device communicates mainly through network infrastructure. | Nodes communicate directly or through multi-hop peer forwarding. |
| Routing | Infrastructure-assisted and centrally coordinated. | Distributed routing among mobile nodes. |
| Mobility | Uses cellular mobility and handoff management. | Uses route discovery and route maintenance as topology changes. |
| Topology | Planned cell structure. | Dynamic topology based on node movement and link availability. |
| Scalability | Can scale through planned site and capacity deployment. | Scalability is affected by routing overhead, node density and wireless contention. |
| Handoff | Handoff between cells or access points is managed by infrastructure. | Connectivity changes as links and routes appear or disappear. |
| Failure handling | Infrastructure failure may affect a large service area. | Alternate peer routes may be possible if enough nodes and links remain. |
| Applications | Public mobile networks, broadband, fleet and healthcare services. | Emergency response, military, temporary and field networks. |

## 3.9 Joint engineering decision: disaster environment

For an emergency or disaster environment in which fixed infrastructure may be unavailable, an ad hoc wireless network is the more suitable initial architecture. Rescue teams, police vehicles, ambulances and portable devices can form a temporary multi-hop network without waiting for base stations or fixed backhaul to be restored.

An ad hoc network can continue operating when nodes remain within communication range of at least one path to the destination. If the network becomes partitioned, communication between separated groups may fail until nodes move or a relay becomes available. Therefore, the design should use redundant routes, mobile relay vehicles and store-and-forward operation where continuous connectivity cannot be guaranteed.

## 3.10 Architecture selection algorithm

```
ALGORITHM SelectWirelessArchitecture(Scenario)

INPUT:
   InfrastructureAvailability
   NodeMobility
   DeploymentTime
   CoverageRequirement
   MultiHopRequirement

OUTPUT:
   Recommended architecture
```

```
BEGIN
  IF InfrastructureAvailability = AVAILABLE AND
    CoverageRequirement = LARGE THEN
     Architecture <- CELLULAR
  ELSE IF DeploymentTime = IMMEDIATE AND
       InfrastructureAvailability = UNAVAILABLE THEN
     Architecture <- AD_HOC
  ELSE IF MultiHopRequirement = REQUIRED THEN
     Architecture <- AD_HOC
  ELSE
     Architecture <- CELLULAR
  END IF

  RETURN Architecture
END
```

## 3.11 Cross-defence

Student A must explain that cellular networks depend on planned base stations, backhaul, core-network services and central coordination. Student B must explain that ad hoc networks can operate without fixed infrastructure because mobile nodes forward packets for one another. The final decision depends on infrastructure availability, coverage requirements, node density, mobility, deployment time and the need for multi-hop communication.

Q4 – APPLICATIONS OF MOBILE IP AND ROUTING REQUIREMENTS

## 4.1 Real-world applications

Mobile IP is useful when a device must maintain network-layer communication while moving between different IP networks. The permanent home address provides a stable identity, while the care-of address represents the current network location. Applications include mobile employees, emergency response, mobile healthcare, transportation and fleet monitoring, field engineering and temporary networks.

## 4.2 Selected application: emergency ambulance communication

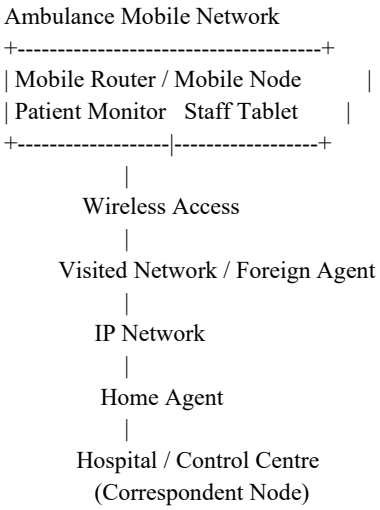
The selected application is an ambulance communication system. Ambulances move between hospitals, accident locations and emergency treatment sites. Medical personnel need to exchange patient information, vital-sign summaries, location updates and operational messages with a hospital control centre. The communication session should continue when the ambulance changes its network attachment.

## 4.3 Mobile entities and communication endpoints

| **Entity** | **Role** |
|---|---|
| Ambulance mobile router / terminal | Provides connectivity for the ambulance equipment and mobile staff. |

| | |
|---|---|
| Mobile medical device | Collects or sends patient monitoring information. |
| Hospital server | Receives patient records, telemetry and status updates. |
| Emergency control centre | Coordinates ambulance location, dispatch and incident response. |
| Home Agent | Maintains the mobile node's home-address-to-care-of-address binding. |
| Foreign Agent / visited network | Supports mobility in traditional Mobile IPv4 deployment. |
| Correspondent Node | Hospital or control-centre endpoint communicating with the mobile node. |

## 4.4 Proposed application architecture




## 4.5 How Mobile IP supports the application

When the ambulance is in its home network, it communicates using its home address. When it enters a visited network, it discovers the network and obtains a care-of address. It registers the care-of address with the home agent. Packets sent by the hospital to the ambulance's home address are intercepted by the home agent and tunnelled to the current care-of address. This allows the application to retain a stable network identity despite movement.

## 4.6 Communication flow

9. The ambulance terminal starts a communication session with the hospital server.
10. The ambulance leaves its home network and enters a visited network.
11. The mobile node discovers the visited network and obtains a care-of address.
12. The mobile node sends a registration request to the home agent.
13. The home agent authenticates the request and updates the binding.
14. The hospital server continues sending packets to the ambulance's home address.
15. The home agent tunnels packets to the registered care-of address.
16. The visited network delivers packets to the ambulance terminal.

17. When the ambulance changes networks again, the care-of address is updated.

## 4.7 Routing requirements

- Stable reachability: the hospital must be able to reach the ambulance despite changes in its point of attachment.
- Low delay: emergency messages and critical patient information should be delivered promptly.
- Reliable delivery: important data should be acknowledged, buffered or retransmitted where appropriate.
- Route maintenance: the network must detect link failure and update routes.
- Security: registration, patient data and operational messages must be authenticated and protected.
- Scalability: the system should support multiple ambulances, medical devices and control-centre endpoints.
- Priority handling: emergency traffic should receive suitable priority over non-critical background traffic.

## 4.8 Delay, packet delivery and interruption

Communication delay can increase during network discovery, authentication, registration and tunnelling. Packet delivery can be interrupted if the old link fails before the new attachment becomes operational. The system should measure handoff interruption, end-to-end delay, packet delivery ratio, packet loss and throughput under different mobility conditions.

## 4.9 Security and reliability

Patient information is sensitive operational data and must be protected from unauthorised access. Mobility registration must be authenticated so that an attacker cannot redirect traffic to an unauthorised care-of address. Communication links should use suitable encryption and access control. Reliability can be improved through redundant connectivity, buffering, acknowledgement and retry mechanisms.

## 4.10 Suitability evaluation

Mobile IP is suitable when the application requires a stable home address and continuity while the mobile node changes IP networks. It is not a complete solution for every communication problem: it does not by itself guarantee radio coverage, sufficient bandwidth, low latency or application-level reliability. It must be combined with reliable wireless access, routing, security and quality-of-service mechanisms.

Student B's primary responsibility is to design the application architecture and explain how Mobile IP supports ambulance mobility. Student A's primary responsibility is to analyse delay, packet delivery, interruption, security and reliability. Both students should be able to justify the selected application and evaluate the complete communication flow.

# Q5 – PROACTIVE ROUTING PROTOCOLS IN MOBILE NETWORKS

## 5.1 Introduction to proactive routing

A proactive routing protocol maintains route information before a source needs to send a packet. Nodes periodically exchange routing information and update their routing tables. When an application generates a packet, a route may already be available, reducing the initial route-discovery delay.

The main disadvantage is control overhead. Periodic updates consume wireless bandwidth and node processing even when routes are not being used. In a highly mobile network, frequent topology changes can cause repeated updates and route-table convergence activity.

## 5.2 DSDV protocol

Destination-Sequenced Distance-Vector (DSDV) is a proactive, table-driven routing protocol for mobile ad hoc networks. It is based on distance-vector routing and uses destination sequence numbers to identify fresh route information. Each node maintains a routing table containing destination, next hop, hop count and sequence number.

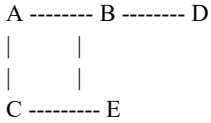
## 5.3 DSDV routing-table structure

| Field | Meaning |
|---|---|
| Destination | Node or network for which a route is maintained. |
| NextHop | Neighbour to which packets should be forwarded. |
| HopCount | Number of links between the current node and destination. |
| SequenceNumber | Freshness value associated with the destination route. |
| RouteState | Indicates whether the route is valid, invalid or unknown. |
| LastUpdate | Time at which the route information was last updated. |

## 5.4 Sequence numbers and route updates

A route with a newer destination sequence number is preferred because it represents fresher information. If two routes have the same sequence number, the route with the smaller hop count can be selected. Sequence numbers help prevent routing loops caused by accepting stale route information. When a link fails, the route can be marked invalid and an updated sequence number can be advertised.

## 5.5 DSDV sample topology




Example:
A reaches D through B.

Initial route: A -> B -> D
Hop count: 2
If link B-D fails, A must receive updated route information.
A may select an alternate path if a valid route exists through C or E.

## 5.6 Route-table example before topology change

| Node | Destination | Next hop | Hop count | Sequence number | State |
|---|---|---|---|---|---|
| A | D | B | 2 | 40 | VALID |
| B | D | D | 1 | 40 | VALID |
| C | D | E | 2 | 40 | VALID |
| E | D | D | 1 | 40 | VALID |

## 5.7 Route update after topology change

Assume the link between B and D fails. Node B detects the failure and marks its route to D invalid. It advertises updated information. Neighbouring nodes compare the received sequence number with their stored route information. A route with a newer sequence number replaces an older route. If an alternative route through C and E is available, A can update its next hop after receiving the fresh information.

| Node | Destination | Next hop | Hop count | Sequence number | State |
|---|---|---|---|---|---|
| A | D | C | 3 | 42 | VALID |
| B | D | — | ∞ | 42 | INVALID |
| C | D | E | 2 | 42 | VALID |
| E | D | D | 1 | 42 | VALID |

The values above are an illustrative desk-check example showing the effect of a link failure and a newer sequence number. They are not measured output from a simulator.

## 5.8 DSDV route-update pseudocode

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:
   Node – receiving mobile node
   ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}

OUTPUT:
   Updated routing table

BEGIN
   Destination <- ReceivedRoute.Destination
   NewHopCount <- ReceivedRoute.HopCount + 1
```

```
    NewSequence <- ReceivedRoute.SequenceNumber

    IF Destination not in RouteTable THEN
       ADD ReceivedRoute with NextHop = Node.Neighbor
    ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
       REPLACE RouteTable[Destination]
    ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
          NewHopCount < RouteTable[Destination].HopCount THEN
       REPLACE RouteTable[Destination]
    END IF

    RETURN RouteTable
END
```

## 5.9 Routing-table maintenance

Each node periodically broadcasts routing information. When an update is received, the node validates it and compares the sequence number and route metric with its current table. Routes that have expired or become invalid are removed or marked unavailable. Periodic and triggered updates help other nodes learn about topology changes.

```
ALGORITHM ProactiveRouteMaintenance(Node)

INPUT:
   Node – mobile node

OUTPUT:
   Updated routing table

BEGIN
   INITIALISE RouteTable

   PERIODICALLY DO
      BROADCAST current routing information
      RECEIVE routing updates from neighbors

      FOR each Update DO
         VALIDATE Update

         FOR each Destination in Update DO
            Calculate NewMetric

            IF NewMetric < CurrentMetric(Destination) OR
               Update.SequenceNumber >
               CurrentSequenceNumber(Destination) THEN
                UPDATE RouteTable[Destination]
                Set NextHop and HopCount
            END IF
         END FOR
      END FOR

      REMOVE expired or invalid entries
   END PERIODICALLY
```

```
    RETURN RouteTable
END
```

## 5.10 Performance investigation

| Metric | Expected DSDV behaviour |
|---|---|
| Route availability | Routes are maintained before data transmission. |
| Initial route delay | Usually low when a valid route is already present. |
| Control overhead | Periodic and triggered updates consume bandwidth. |
| Convergence | Time is required for updated information to propagate after topology change. |
| Mobility impact | Frequent movement causes more route updates and possible stale information. |
| Network size | Larger networks can increase routing-table size and update traffic. |

## 5.11 Evaluation

DSDV is suitable when nodes communicate frequently and low route-discovery delay is important. It may be less efficient when traffic is intermittent and the topology changes frequently, because periodic updates are transmitted even when routes are not currently needed. Its performance should be evaluated using control-message count, route availability, convergence time, end-to-end delay and packet delivery ratio.

Student A's primary responsibility is to explain proactive routing, DSDV, sequence numbers, routing-table maintenance and the sample topology. Student B's primary responsibility is to measure or analyse routing overhead, route availability, convergence, control messages, delay and packet delivery.

Q6 – REACTIVE ROUTING AND PROTOCOL COMPARISON

## 6.1 Introduction to reactive routing

A reactive routing protocol discovers a route only when a source needs to communicate with a destination and no valid route is available. This is called on-demand routing. It avoids maintaining every possible route continuously, reducing periodic control traffic when communication is infrequent. However, the first packet may experience route-discovery delay.

## 6.2 AODV protocol

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. It discovers routes when required and maintains active routes while they are being used. AODV uses Route Request (RREQ),

Route Reply (RREP) and Route Error (RERR) messages. Destination sequence numbers help select fresh routes and reduce routing loops.

## 6.3 Route Request (RREQ)

When a source has data to send but does not have a valid route, it broadcasts an RREQ. The request contains source and destination information, a request identifier and routing information such as hop count and sequence numbers. Intermediate nodes record a reverse route toward the source and forward the request if it has not already been processed.

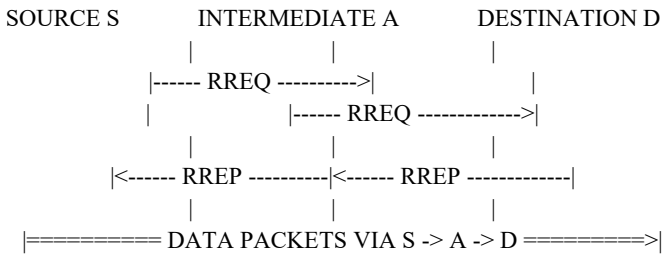
## 6.4 Route Reply (RREP)

When the destination receives an RREQ, it creates an RREP and sends it back along the reverse path. An intermediate node may also reply if it has a sufficiently fresh valid route to the destination. As the RREP travels back, nodes establish the forward route toward the destination.

## 6.5 Route Error (RERR)

When an active link breaks and no valid alternate route is available, a node generates an RERR message to notify affected nodes that a destination is no longer reachable through that route. The source may then initiate a new route discovery if communication is still required.

## 6.6 AODV route discovery sequence




## 6.7 AODV route-request pseudocode

```
ALGORITHM ProcessRouteRequest(Node, Request)

INPUT:
   Node – current receiving node
   Request = {Source, Destination, RequestID, HopCount}

OUTPUT:
   Route discovery action

BEGIN
   IF Request already processed THEN
      DISCARD Request
      RETURN
   END IF

   Record reverse route toward Request.Source
   Mark Request as processed
```

```
    IF Node = Request.Destination THEN
      Create RouteReply
      SEND RouteReply toward Source
      RETURN
    END IF

    IF ValidRouteTo(Request.Destination) THEN
      Create RouteReply using valid route
      SEND RouteReply toward Source
    ELSE
      Request.HopCount <- Request.HopCount + 1
      FOR each Neighbor in NeighborTable DO
        FORWARD Request TO Neighbor
      END FOR
    END IF
END
```

## 6.8 Route maintenance pseudocode

```
ALGORITHM MaintainReactiveRoute(Source, Destination)

INPUT:
  Source and Destination

OUTPUT:
  Route status

BEGIN
  MONITOR active route

  IF LinkFailureDetected THEN
    Mark failed link as INVALID

    IF AlternateValidRoute exists THEN
      Select alternate route
      RETURN ALTERNATE_ROUTE_USED
    ELSE
      Generate RouteError
      SEND RouteError toward affected source

      IF Source still requires communication THEN
        INITIATE new RouteDiscovery
      ELSE
        RETURN ROUTE_TERMINATED
      END IF
    END IF
  END IF

  RETURN ROUTE_ACTIVE
END
```

## 6.9 DSDV and AODV comparison

| Feature | DSDV – Proactive | AODV – Reactive |
|---|---|---|
| Route availability | Maintained before data transmission. | Obtained when required. |
| Route discovery delay | Usually low if route is already valid. | Present when route is unknown. |
| Control overhead | Periodic and triggered updates. | Demand-driven discovery and maintenance. |
| Routing table | Maintains destination routes continuously. | Maintains routes needed for active communication. |
| Topology changes | Updates routing tables as changes are learned. | Repairs active routes or starts new discovery. |
| Intermittent traffic | May spend bandwidth maintaining unused routes. | Often efficient because routes are discovered on demand. |
| Frequent traffic | Can provide quick forwarding when routes are stable. | Repeated route discovery may be avoided while route remains valid. |
| Example use | Networks with regular traffic and relatively stable topology. | Networks with intermittent traffic and changing topology. |

## 6.10 Selection for a highly dynamic mobile network

For a highly dynamic mobile network with frequent topology changes, AODV is generally the more suitable choice when traffic is intermittent or when maintaining every route continuously would create excessive control overhead. It discovers routes when needed and repairs active routes when a link fails. However, AODV can experience route-discovery delay and repeated RREQ broadcasts when mobility is very high.

DSDV may be preferred when communication is frequent and low initial route delay is critical, provided that the network can tolerate periodic routing updates and the topology is not changing too rapidly. The final selection must consider traffic rate, mobility rate, available bandwidth, delay requirements and network size.

## 6.11 Decision algorithm

```
ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate,
                    DelayRequirement, Bandwidth)

INPUT:
   TrafficRate – frequency of data communication
   MobilityRate – frequency of topology changes
   DelayRequirement – acceptable route setup delay
   Bandwidth – available bandwidth

OUTPUT:
   Recommended routing approach

BEGIN
```

```
    IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN
       Recommendation <- PROACTIVE
    ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN
       Recommendation <- REACTIVE
    ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN
       Recommendation <- REACTIVE
    ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN
       Recommendation <- PROACTIVE
    ELSE
       Recommendation <- EVALUATE_BOTH
    END IF

    RETURN Recommendation
END
```

## 6.12 Joint conclusion

AODV is selected for a highly dynamic network with intermittent traffic because it avoids continuous route maintenance for destinations that are not being used. Its disadvantage is route-discovery delay. DSDV is more suitable for regular traffic and situations where a route should be available immediately. A network designer should compare routing overhead, delay, packet delivery, route stability and mobility rather than selecting a protocol based on one metric alone.

Student A's primary responsibility is reactive routing and AODV operation. Student B's primary responsibility is comparative verification of AODV and DSDV. Both students must explain the two protocols and justify a selection under changed network conditions.

Q7 – DESIGN A MOBILE EMERGENCY COMMUNICATION NETWORK

## 7.1 Scenario and design objectives

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may be damaged or unavailable. Emergency teams must exchange critical information with minimum delay. The network must therefore be rapidly deployable, support mobility, tolerate link and node failures and continue operating when some network segments become disconnected.

## 7.2 Stage 1 – Independent design proposal: Student A

### Wireless architecture

Student A proposes a hybrid emergency architecture. Mobile rescue units, ambulances, police vehicles and portable team devices form an ad hoc multi-hop network. Where cellular infrastructure remains available, the mobile nodes may also use it as an additional access path. Mobile relay vehicles or portable access points can connect separated groups and provide gateway connectivity to the emergency control centre.

### Mobility-management strategy

Mobile IP is used where continuity of communication across different IP networks is required. Each mobile node retains a home address and registers a care-of address when it enters a visited network. When infrastructure is unavailable and no home-agent path can be reached, local ad hoc communication continues using the ad hoc routing protocol. Mobile IP therefore supports IP mobility when its agents and registration path are reachable; it does not replace ad hoc routing.

### Communication model

Emergency messages are prioritised. Short critical messages, such as distress alerts and patient status, are sent ahead of non-critical bulk data. Nodes maintain local communication with nearby peers and forward packets over multiple hops toward the control centre or a relay gateway. If no path exists, the device stores the message and forwards it when connectivity returns.

## 7.3 Stage 1 – Independent design proposal: Student B

### Routing strategy

Student B proposes AODV as the primary ad hoc routing protocol. AODV discovers a route when a source needs to communicate and maintains the route while it is active. This reduces periodic routing traffic when many possible destinations are not being used. RREQ and RREP establish paths, while RERR messages report broken routes.

### Route maintenance and failure handling

- Monitor active links and mark a route invalid when a link failure is detected.
- Use an alternate valid route when one is available.
- Generate and propagate RERR when an active route cannot be repaired locally.
- Start a new route discovery if the source still requires communication.
- Use mobile relay nodes to bridge network partitions where possible.
- Buffer critical packets during temporary disconnection and forward them when a route becomes available.

### Performance parameters

Student B proposes measuring end-to-end delay, packet delivery ratio, packet loss ratio, routing-control overhead, route-discovery delay, route recovery time and network availability. Tests should vary node count, mobility speed, link failures and traffic load.

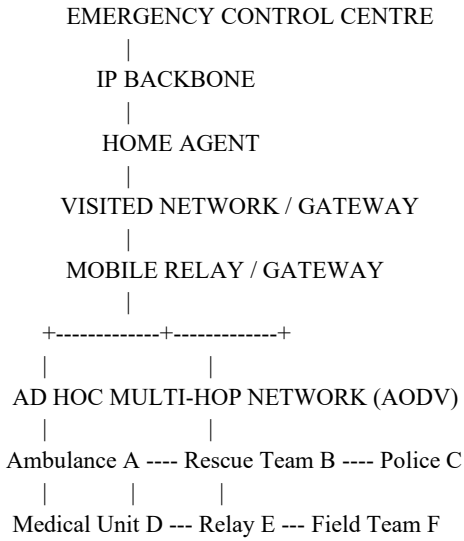
## 7.4 Stage 2 – Compare the two designs

| Design aspect | Student A proposal | Student B proposal | Joint interpretation |
|---|---|---|---|
| Architecture | Hybrid cellular and ad hoc access. | Distributed ad hoc network. | Use ad hoc as the resilient local layer and cellular where available. |
| Mobility | Mobile IP for stable IP identity across visited | Route discovery and maintenance as nodes | Mobile IP manages IP location; AODV manages |

| | | | |
|---|---|---|---|
| | networks. | move. | multi-hop routes. |
| Routing | Mobility-aware forwarding and infrastructure support. | AODV as primary on-demand routing. | Use AODV for local ad hoc forwarding. |
| Failure handling | Use alternate access and temporary gateways. | Alternate routes, RERR and rediscovery. | Combine redundant access, route repair and store-and-forward. |
| Performance | Focus on registration, tunnel overhead and interruption. | Focus on route discovery, delay and packet delivery. | Measure both mobility and routing overhead. |

## 7.5 Stage 3 – Joint final architecture

The final design uses a hybrid emergency communication network. Ad hoc nodes provide local multi-hop communication when fixed infrastructure is unavailable. Cellular or other surviving infrastructure is used when available to reach remote services. Portable gateways and mobile relay vehicles connect ad hoc segments to the emergency control centre. Mobile IP supports stable communication for nodes that move between IP networks and can reach their home agent.




Each mobile node can send, receive and forward packets.
Surviving cellular access may provide an additional uplink.

## 7.6 Mobile IP components in the final design

| **Mobile IP component** | **Emergency-network role** |
|---|---|
| Mobile Node | Ambulance terminal, rescue vehicle router or mobile command device. |
| Home Agent | Maintains the home-address-to-care-of-address binding for the mobile node. |
| Foreign Agent | Assists with mobility in a traditional Mobile IPv4 visited network. |

| | |
|---|---|
| Correspondent Node | Emergency control centre, hospital server or dispatch system. |
| Care-of Address | Represents the current visited-network attachment. |

## 7.7 Registration and tunnelling process

18. A mobile emergency unit detects a visited IP network.
19. It discovers the mobility agent or obtains a care-of address.
20. It sends a registration request containing its home address and care-of address.
21. The home agent authenticates the request and updates the binding.
22. The correspondent node sends packets to the mobile node's home address.
23. The home agent encapsulates and tunnels packets to the care-of address.
24. The visited network or foreign agent decapsulates and delivers the packets.
25. When the mobile node changes networks, it obtains a new care-of address and updates the registration.

## 7.8 Routing protocol selection

AODV is selected for the local ad hoc layer because disaster networks can have intermittent traffic and frequent topology changes. It does not require every node to maintain a complete route table for all destinations continuously. When an emergency message must be sent and no valid route exists, AODV initiates route discovery. If a link breaks, the route is repaired or an RERR is sent and a new route may be discovered.

AODV has limitations. Route discovery can add delay, and frequent movement can trigger repeated RREQ broadcasts. For stable high-rate communication where very low route setup delay is essential, a proactive protocol such as DSDV may be useful in a smaller, relatively stable subnetwork. The final design can therefore evaluate both approaches, but AODV remains the primary choice for the highly dynamic multi-hop layer.

## 7.9 Failure handling

| **Failure condition** | **Response** |
|---|---|
| Mobile node failure | Remove or expire routes through the failed node; use another route if available. |
| Wireless link failure | Mark the link down, invalidate affected route and attempt alternate route or rediscovery. |
| Gateway failure | Select another gateway or relay if one is reachable. |
| Home Agent unreachable | Continue local ad hoc communication; queue remote-bound packets until mobility services become reachable. |
| Network partition | Use store-and-forward, mobile relays and delayed delivery when connectivity returns. |
| Congestion | Prioritise emergency traffic and reduce or defer non-critical data. |

| | |
|---|---|
| Security attack | Authenticate nodes and mobility messages; reject unauthorised registration and routing information. |

## 7.10 Stage 4 – Analytical validation by Student A

### Mobile IP overhead

Mobile IP adds registration messages and encapsulation overhead. The registration exchange consumes signalling bandwidth and processing time whenever the mobile node changes its care-of address. Tunnelling adds an outer IP header to each encapsulated packet. The total overhead depends on packet size, registration frequency and the number of mobile nodes.

### Mobility-related delay

Mobility delay includes network discovery, care-of address acquisition, registration and route convergence. During this interval, packets may be delayed or lost. Frequent movement increases the number of mobility updates. The design should reduce unnecessary registration and coordinate handoff with packet forwarding.

### Packet delivery and interruption

Packet delivery depends on whether a valid path exists from the source to the destination. AODV route discovery can temporarily delay delivery when no route is known. Mobile IP can maintain reachability across visited networks when the home agent and tunnel are reachable. During a complete partition, neither mechanism can deliver packets immediately; buffering and later forwarding are required.

## 7.11 Stage 4 – Experimental validation by Student B

The experimental validation should use a small network simulation or a reproducible testbed. The assignment brief does not provide measured experimental outputs, so the following are proposed test cases and result-recording templates. Actual values must be filled in after the test is run.

| Test case | Nodes | Mobility condition | Failure condition | Purpose |
|---|---|---|---|---|
| T1 | 10 | Low mobility | No planned failure | Baseline delay and delivery. |
| T2 | 20 | Moderate mobility | One link failure | Observe route maintenance. |
| T3 | 30 | High mobility | Repeated link changes | Measure route discovery and overhead. |
| T4 | 40 | High mobility | Gateway failure | Evaluate alternate gateway and recovery. |
| T5 | 30 | Intermittent movement | Network partition | Evaluate buffering and delayed |

| | | | | |
|---|---|---|---|---|
| | | | | forwarding. |

| Metric | How to calculate / record |
|---|---|
| Average end-to-end delay | Total packet delivery time divided by number of packets received. |
| Packet delivery ratio | Received data packets / generated data packets × 100. |
| Packet loss ratio | Lost data packets / generated data packets × 100. |
| Routing overhead | Routing control packets transmitted, or control bytes divided by delivered data bytes. |
| Route discovery delay | Time from RREQ initiation to successful route establishment. |
| Recovery time | Time from link failure detection to successful data forwarding over a valid route. |

## 7.12 Result recording template

| Scenario | Average delay | PDR (%) | Routing overhead | Recovery time | Observation |
|---|---|---|---|---|---|
| Low mobility | To be measured | To be measured | To be measured | To be measured | Baseline. |
| Moderate mobility | To be measured | To be measured | To be measured | To be measured | Observe single link failure. |
| High mobility | To be measured | To be measured | To be measured | To be measured | Observe repeated route changes. |
| Gateway failure | To be measured | To be measured | To be measured | To be measured | Observe gateway recovery. |
| Network partition | To be measured | To be measured | To be measured | To be measured | Observe store-and-forward. |

The result table intentionally uses 'To be measured' rather than invented values. A valid experimental report should state the simulator or testbed, network area, radio range, traffic model, packet size, simulation duration, mobility model, random seed and number of repetitions.

## 7.13 Stage 5 – Cross-defence

Student A must explain why AODV is appropriate for the emergency network. AODV is demand-driven and avoids continuous route maintenance for every possible destination. It can discover routes when needed and report broken routes through RERR. This is useful when traffic is intermittent and topology changes frequently. Its route-discovery delay and control-message bursts must be considered.

Student B must explain how Mobile IP maintains communication when a mobile node changes network location. The mobile node obtains a new care-of address and registers it with the home agent.

The home agent updates the binding and tunnels packets addressed to the home address toward the current care-of address. This preserves the stable home address while the point of attachment changes.

## 7.14 Stage 6 – Joint evaluation

- Architecture suitability: the hybrid design combines rapid ad hoc deployment with surviving infrastructure where available.
- Mobile IP requirement: it is useful for maintaining IP-level reachability across visited networks, but it is not required for every local peer-to-peer ad hoc exchange.
- Routing selection: AODV is appropriate for the dynamic multi-hop layer because it discovers routes on demand.
- Network overhead: registration, tunnelling, RREQ, RREP and RERR messages consume bandwidth and processing resources.
- Mobility impact: movement changes links, routes and care-of addresses, which may cause delay and packet loss.
- Connectivity loss: when no path exists, packets cannot be delivered immediately; buffering, store-and-forward and mobile relays can help.
- Limitations: the design may not meet strict real-time requirements in a heavily partitioned network or where node density and radio range are insufficient.

## 7.15 Final conclusion for the design challenge

Before selecting Mobile IP and a routing protocol for a highly mobile wireless environment, a network designer should analyse infrastructure availability, node density, movement speed, traffic frequency, bandwidth, end-to-end delay requirements, packet delivery, security, energy constraints, route stability and expected network partitions. Mobile IP should be selected when stable IP identity across different networks is required and the home agent can be reached. AODV is suitable for dynamic ad hoc routing with intermittent traffic, while DSDV may be considered where routes must be available immediately and the topology is sufficiently stable.

The proposed emergency network combines ad hoc multi-hop communication, Mobile IP mobility support where infrastructure is reachable, AODV route discovery and repair, redundant gateways, mobile relays and store-and-forward operation. This approach provides a flexible response to damaged infrastructure while recognising that no routing protocol can deliver packets across a complete network partition without a communication path.

CONTRIBUTION RECORD

The assignment brief requires both students to participate in every question and understand the complete submission. The following table follows the contribution structure provided in the question paper. The partner's name is not included because it was not supplied.

| **Question** | **Student A – Primary contribution** | **Student B – Primary contribution** | **Joint work** |
|---|---|---|---|
| Q1 | Network architecture analysis | Mobility analysis | Wireless network analysis |

| | | | |
|---|---|---|---|
| Q2 | Mobile IP operation | Mobile IP challenges | Mobile IP architecture |
| Q3 | Cellular network investigation | Ad hoc network investigation | Comparative analysis |
| Q4 | Performance and requirement analysis | Application architecture | Application evaluation |
| Q5 | Protocol design | Performance investigation | Proactive routing |
| Q6 | Reactive routing | DSDV–AODV comparison | Routing evaluation |
| Q7 | Analytical validation | Experimental validation | Final emergency network design |

# INDIVIDUAL VERIFICATION – VIVA PREPARATION

## Q1 – Wireless network architecture

- What are the main components of a wireless network?
- Why does limited bandwidth affect the number of active users?
- What is a handoff and why can it cause packet loss?
- How does a mobile node select a new access point?

## Q2 – Mobile IP

- What is the difference between a home address and a care-of address?
- What is the role of the Home Agent?
- What is tunnelling in Mobile IP?
- What is triangular routing and why can it increase delay?
- How does registration update the mobile node's location?

## Q3 – Cellular and ad hoc networks

- Why does a cellular network depend on base stations?
- How can an ad hoc network communicate without fixed infrastructure?
- What is multi-hop communication?
- Which architecture is suitable when fixed infrastructure is unavailable?

## Q4 – Mobile IP applications

- Why is Mobile IP useful for an ambulance communication system?
- What happens when an ambulance enters a visited network?
- What are the security requirements for mobile healthcare data?
- Does Mobile IP guarantee low delay or continuous radio coverage?

### Q5 – Proactive routing

- What is proactive routing?
- Why does DSDV use sequence numbers?
- What information is stored in a DSDV routing table?
- What is the main disadvantage of periodic route updates?

### Q6 – Reactive routing

- What is on-demand route discovery?
- What are RREQ, RREP and RERR?
- How does AODV respond to a broken link?
- When would DSDV be preferred over AODV?

### Q7 – Emergency network design

- Why is a hybrid ad hoc and infrastructure architecture useful in a disaster?
- How does Mobile IP maintain communication when a mobile node changes networks?
- Why was AODV selected for the ad hoc layer?
- What happens when the network becomes partitioned?
- Which performance metrics should be measured during validation?

## OVERALL CONCLUSION

Wireless networks support mobile communication through radio links, access points, base stations and mobile nodes. Mobility introduces changing attachment points, variable signal quality, handoff delay, packet loss and route instability. A suitable wireless architecture must consider bandwidth, connectivity, coverage, mobility and application requirements.

Mobile IP supports network-layer mobility by preserving a mobile node's home address and maintaining a binding to its current care-of address. The Home Agent forwards packets through a tunnel toward the visited network. Registration delay, tunnelling overhead, triangular routing, packet loss, security and scalability must be considered when designing a mobile system.

Cellular networks provide planned infrastructure and wide-area coverage, while ad hoc networks allow rapid infrastructure-less deployment through multi-hop forwarding. Proactive routing such as DSDV maintains routes continuously and can reduce initial route-discovery delay, but it introduces periodic control overhead. Reactive routing such as AODV discovers routes when required and is useful for intermittent traffic and dynamic topologies, although route discovery adds delay.

For the emergency communication scenario, a hybrid architecture is recommended. Ad hoc networking provides local communication when fixed infrastructure is unavailable, Mobile IP supports stable IP reachability where home-agent connectivity exists, and AODV provides on-demand multi-hop routing. Redundant gateways, mobile relays and store-and-forward operation improve resilience. The design must be validated using measured delay, packet delivery ratio, routing overhead and recovery time under different mobility and failure conditions.

# REFERENCES


- ITA03 – Mobile Computing, Assignment 3 – CO3, supplied assignment brief, pages 1–12.
- SIMATS Engineering, Mobile Computing: Standard for Algorithmic Pseudocode, supplied assignment handbook, pages 13–40.
- Mobile IP terminology and operation as defined in the supplied assignment handbook: Mobile Node, Home Agent, Foreign Agent, Care-of Address and Correspondent Node.
- Routing protocol terminology and procedures as defined in the supplied assignment handbook: DSDV-style proactive routing and AODV-style reactive routing.